

Layered Structure of the TCP/IP Model

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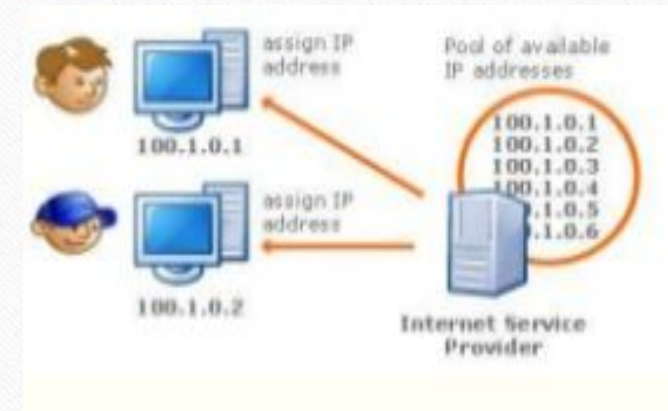
What is Protocol, IP?

Protocol-

In Information Technology, a **protocol** is the special set of rules that end points in a telecommunication connection use when they communicate. Protocols specify interactions between the communicating entities.

IP-

An Internet Protocol address(**IP** address) is a numerical label assigned to each device(e.g., computer, printer, etc.) participating in a computer network that uses the Internet Protocol for communication.

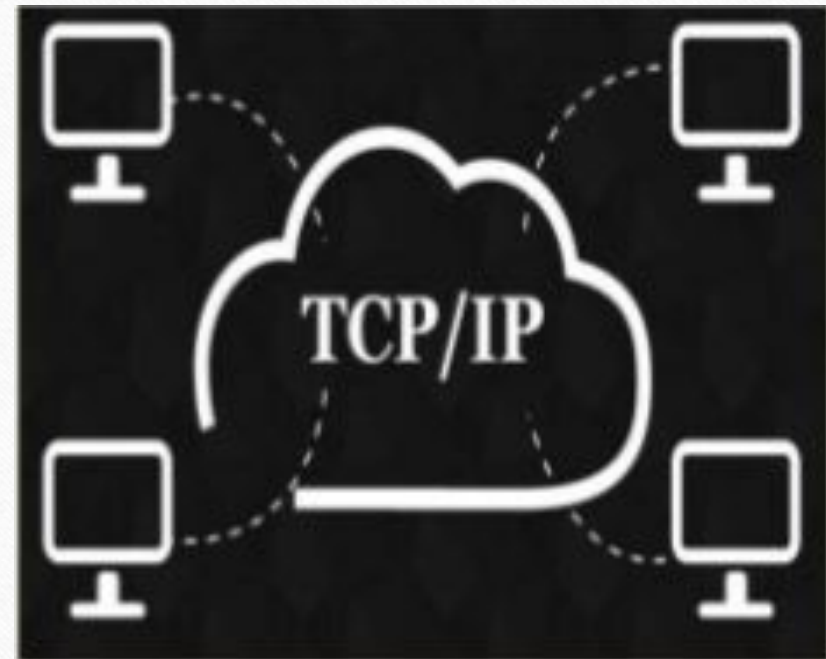


What is the tcp/ip model?

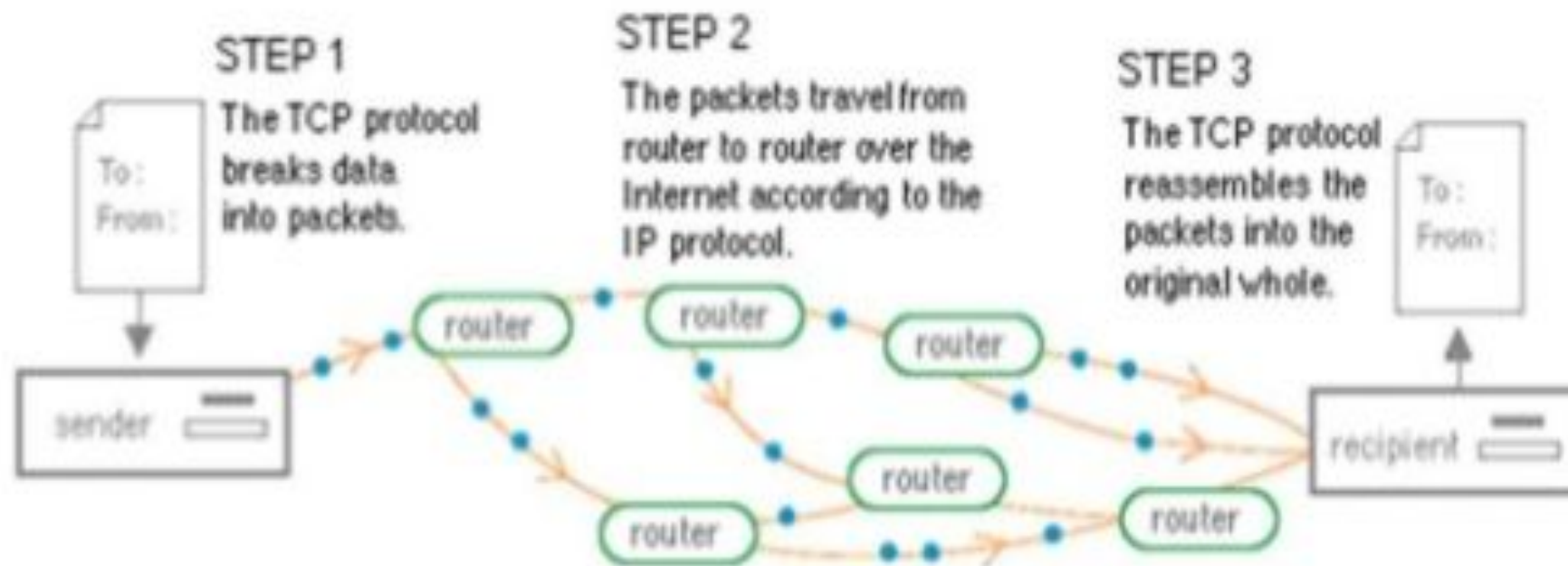
- **TCP/IP Model** helps you to determine how a specific computer should be connected to the Internet and how data should be transmitted between them.
- TCP/IP stands for **Transmission Control Protocol/Internet Protocol**.
- It was designed by the **Advanced Research Project Agency Network[ARPANET]** which is one of the famous research network sponsored by the DoD(U.S. Department of Defense).
- TCP/IP helps you to create a virtual network when multiple computer networks are connected together.
- The purpose of TCP/IP model is to allow communication over a long distances.

History Of the TCP/IP model:

- The Transmission Control Protocol/Internet Protocol suite of protocol from the basis of the Internet.
- It is TCP/IP that creates a virtual network when multiple computer network are connected together.
- The TCP/IP networks was earlier known as ARPANET, but is now known as Internet.



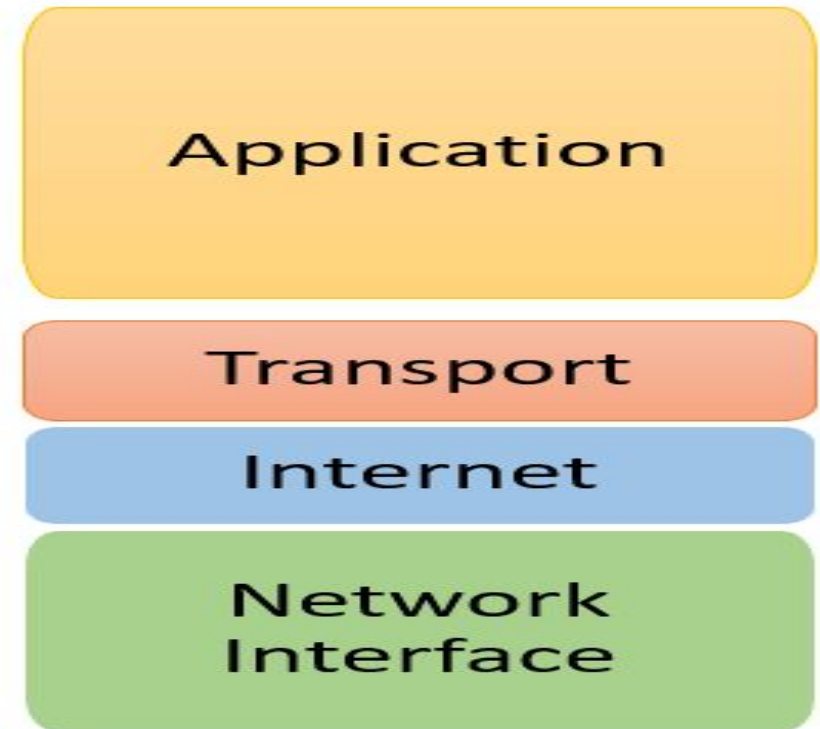
How tcp/ip model works:



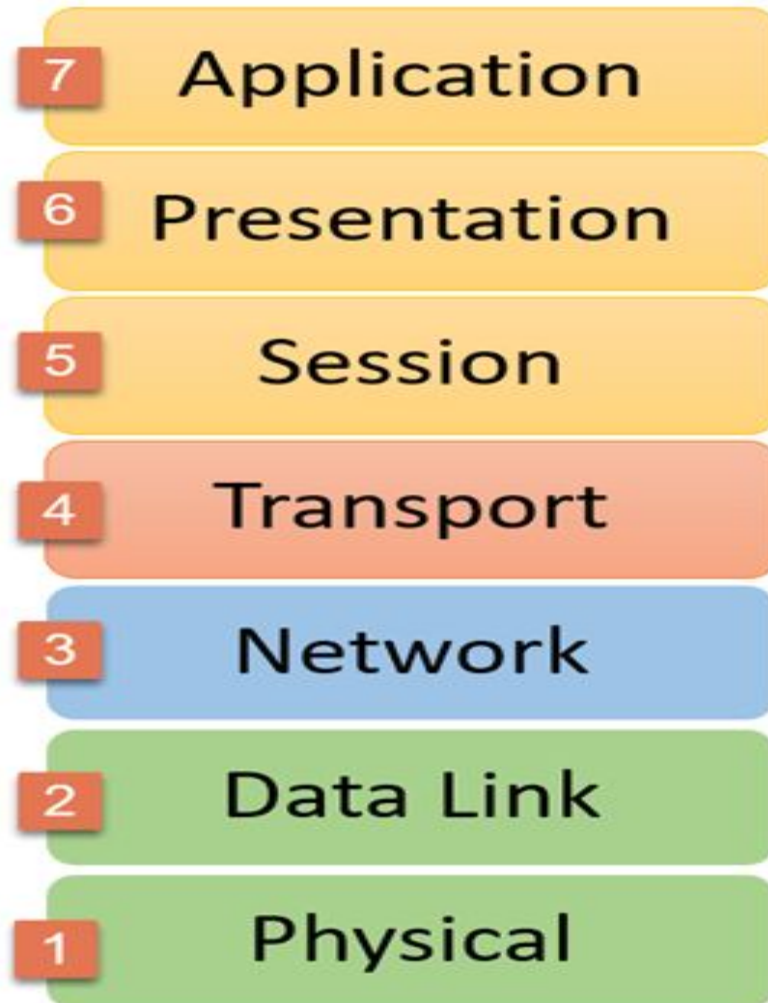
Layered structure of TCP/IP Model:

❑ Four Layers Of TCP/IP Model:

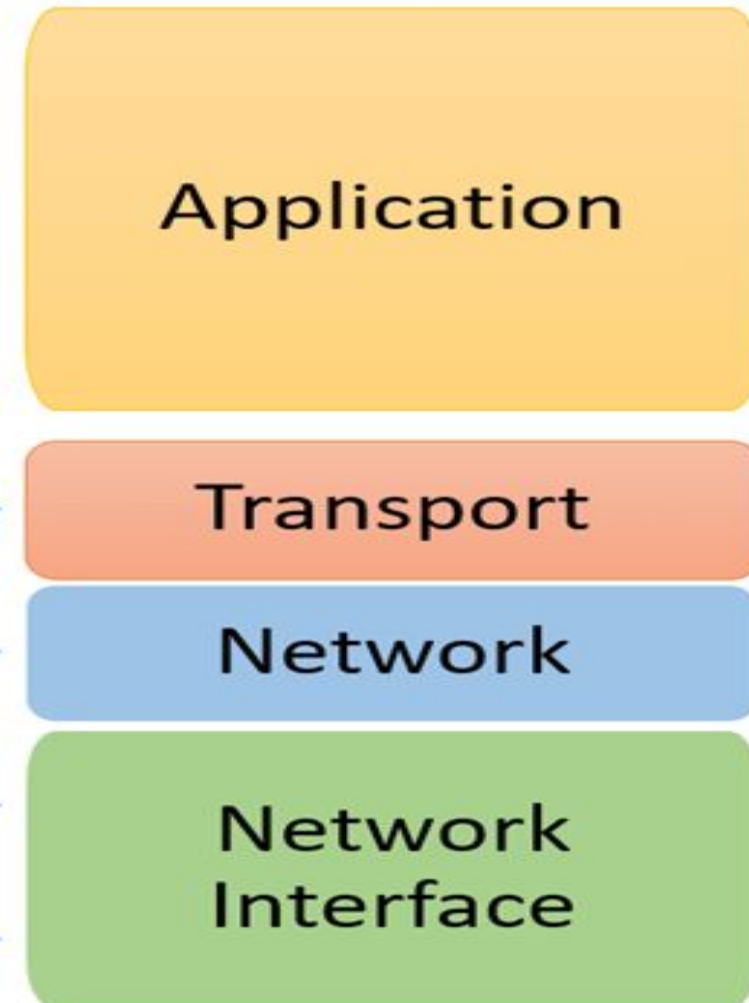
1. Network Interface Layer(also known as Host-to-Network Layer)
2. Internet Layer
3. Transport Layer
4. Application Layer



OSI Reference Model



TCP/IP Conceptual Layers



1. NETWORK INTERFACE LAYER:

- ❑ Network Interface Layer is the very first layer of the four-layered TCP/IP model which corresponds to the Data Link and Physical Layer of the OSI Reference Model.
- ❑ It is also known as Host-to-Network Layer.
- ❑ It helps us to defines the details of how data should be sent using the network.
- ❑ Network Interface Layer combines raw data into data frames.
- ❑ Provides physical interface for data transmission.
- ❑ Standard Ethernet, Token-ring, Point-to-Point Protocol (PPP), are the protocols which are used in this Layer.

Functions of NETWORK INTERFACE LAYER:

- The main function of the Network Interface Layer is to connect the Host to the Network and inform the upper layers so that they could start sending the data packets.
- Three Important Functions:
 - 1.**Path Determination**- It determines the route taken by packets from source to destination.
 - 2.**Forwradng**- Move packets from router's input to the appropriate router output.
 - 3.**Call Setup**- Some network architectures require router call setup along path before data flows.

2. Internet LAYER:

- ❑ Internet Layer is the second layer of the TCP/IP model to the top of the Network Interface Model.
- ❑ The Internet Layer is exactly same as that of Network Layer of OSI reference model.
- ❑ The main work of this layer is to send the packets from any network, and any computer still they reach the destination irrespective of the route they take.
- ❑ It defines protocols which are responsible for logical transmission of data over entire network.

IP Protocol:

IP is the primary protocol operating at this layer and it provides data routing, addressing and fragmentation services to the protocols at the transport layer about it.

Functions of Internet LAYER:

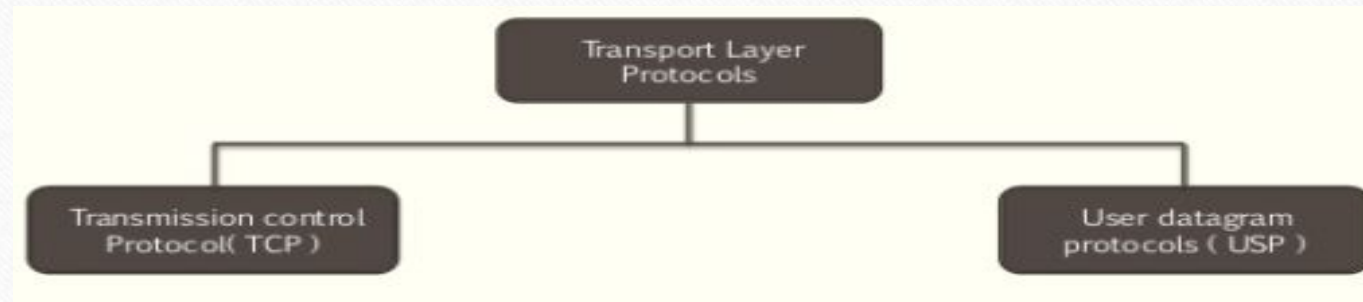
- It transmits data packets to the Network Interface Layer.
- It routes each of the data packets independently from the source to destination.
- It reassembles the out-of-order packets when they reach the destination.
- It handles the error in transmission of data packets and fragmentation of data packets.



3. Transport LAYER:

- The Transport Layer is the third layer of the TCP/IP model which builds on the network layer in order to provide data transport from a process on a source system to a process on a destination system.
- It determines how much data should be sent where and at what rate.
- Transport layer helps you to control the reliability of a link through flow control, error control, and segmentation or de-segmentation .Hence, it ensure that data units are delivered error-free and in sequence.
- The transport layer also offers an acknowledgment of the successful data transmission and sends the next data in case no errors occurred.

Types of Protocols used in transport layer:



Parameter	UDP	TCP
Data Transfer	Data is sent in discrete packages by the application.	Data is sent by the application with no particular structure.
Transmission speed	Very High	High but not as high as UDP
Protocol connection setup	Connectionless	Connection oriented
Used	UDP is useful when speed of delivery is critical	TCP is useful for transmission of data without error.

Functions of Transport LAYER:

- It **divides** the message received from the session layer into segments and numbers them to make a sequence.
- Transport layer makes sure that the message is delivered to the correct process on the destination machine.
- It also makes sure that the entire message arrives without any error else it should be retransmitted.

4. application LAYER:

- ❑ Application Layer is the topmost layer of TCP/IP model which performs the functions of top three layers of the OSI model: Application, Presentation and Session Layer.
- ❑ Application layer interacts with the Application Programs which are closer to the user.
- ❑ It is responsible for node-to-node communication and controls user-interface specifications.
- ❑ It provides a way for applications to have access networked services.

Types of Protocols in application layer:

1. **TELNET**- TELNET is the virtual terminal protocol which allows a user on one machine to log onto a distant machine and work there.
2. **FTP**- File Transfer Protocol provides a way to move data efficiently from one machine to another.
3. **HTTP**- Hypertext Transfer Protocol (HTTP) is used for fetching pages on the World Wide Web(www) and many others.



Functions of application LAYER:

- Application-layer helps you to identify communication partners, determining resource availability, and synchronizing communication.
- It allows users to log on to a remote host
- This layer provides various e-mail services
- This application offers distributed database sources and access for global information about various objects and services.

Advantages of TCp/ip model:

- It helps you to establish/set up a connection between different types of computers.
- It operates independently of the operating system.
- It supports many routing-protocols.
- It enables the internetworking between the organizations.
- TCP/IP model has a highly scalable client-server architecture.
- It can be operated independently.

disadvantages of TCp/ip model:

- TCP/IP is a complicated model to set up and manage.
- The shallow/overhead of TCP/IP is higher-than IPX (Internetwork Packet Exchange).
- In this, model the transport layer does not guarantee delivery of packets.
- Replacing protocol in TCP/IP is not easy.
- It has no clear separation from its services, interfaces, and protocols.

Summary of the layers of TCp/ip model:

Application

- To allow access to network resources

Transport

- To provide reliable process to process message delivery and error delivery

Internet

- To move packets from source to destination
- To provide internetworking

Network Interface

Responsible for the transmission for the between two device on the same network.



COMPARISON BETWEEN TCP/IP AND OSI

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What is OSI Model?

- ▣ The OSI Model is a logical and conceptual model that defines network communication used by systems open to interconnection and communication with other systems. The Open System Interconnection (OSI Model) also defines a logical network and effectively describes computer packet transfer by using various layers of protocols.

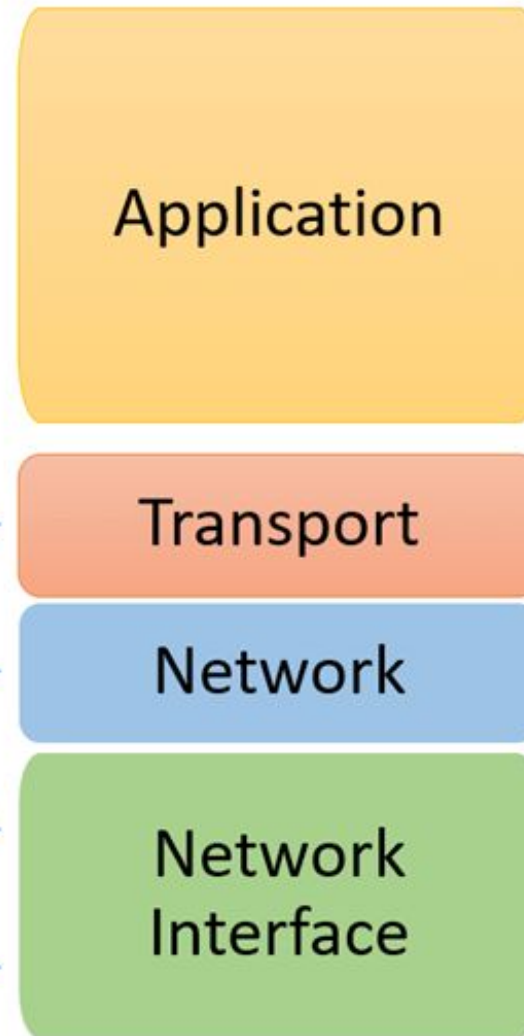
What is TCP/IP Model?

- ▣ TCP/IP helps you to determine how a specific computer should be connected to the internet and how you can transmit data between them. It helps you to create a virtual network when multiple computer networks are connected together.
- ▣ TCP/IP stands for Transmission Control Protocol/Internet Protocol. It is specifically designed as a model to offer highly reliable and end-to-end byte stream over an unreliable internetwork.

OSI Reference Model



TCP/IP Conceptual Layers



KEY DIFFERENCE

- ❑ OSI has 7 layers whereas TCP/IP has 4 layers.
- ❑ The OSI Model is a logical and conceptual model that defines network communication used by systems open to interconnection and communication with other systems. On the other hand, TCP/IP helps you to determine how a specific computer should be connected to the internet and how you can be transmitted between them.
- ❑ OSI header is 5 bytes whereas TCP/IP header size is 20 bytes.
- ❑ OSI refers to Open Systems Interconnection whereas TCP/IP refers to Transmission Control Protocol.
- ❑ OSI follows a vertical approach whereas TCP/IP follows a horizontal approach.
- ❑ OSI model, the transport layer, is only connection-oriented whereas the TCP/IP model is both connection-oriented and connectionless.
- ❑ OSI model is developed by ISO (International Standard Organization), whereas TCP Model is developed by ARPANET (Advanced Research Project Agency Network).
- ❑ OSI model helps you to standardize router, switch, motherboard, and other hardware whereas TCP/IP helps you to establish a connection between different types of computers.

Comparison

OSI Model	TCP/IP Model
It is developed by ISO (International Standard Organization)	It is developed by ARPANET (Advanced Research Project Agency Network).
OSI model provides a clear distinction between interfaces, services, and protocols.	TCP/IP doesn't have any clear distinguishing points between services, interfaces, and protocols.
OSI refers to Open Systems Interconnection.	TCP refers to Transmission Control Protocol.
OSI uses the network layer to define routing standards and protocols.	TCP/IP uses only the Internet layer.
OSI follows a vertical approach.	TCP/IP follows a horizontal approach.
OSI layers have seven layers.	TCP/IP has four layers.
In the OSI model, the transport layer is only connection-oriented.	A layer of the TCP/IP model is both connection-oriented and connectionless.
In the OSI model, the data link layer and physical are separate layers.	In TCP, physical and data link are both combined as a single host-to-network layer.
Session and presentation layers are a part of the OSI model.	There is no session and presentation layer in the TCP model.
It is defined after the advent of the Internet.	It is defined before the advent of the internet.
The minimum size of the OSI header is 5 bytes.	The minimum header size is 20 bytes.



**thank
you!**



UNIT 5

OSI REFERENCE MODEL & TCP/IP MODEL

PRESENTED BY:

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CONTENT:

- 5.4 IP Addressing
- 5.5 Introduction to Dynamic Host Configuration Protocol(DHCP) and Network Address Translation (NAT)
- 5.6 Ethernet
- 5.7 Virtual LAN(VLAN) ,Fast and Gigabit Ethernet, Token Ring

IP ADDRESSING

- IP Address with classes
- IP Address with classless IP Addressing
- Subnetting
- Supernetting, Masking
- Introduction to IPV6 address.

WHAT IS IP?

- An IP (Internet Protocol) address is a numerical label assigned to the devices connected to a computer network that uses the IP for communication.
- It also helps you to develop a virtual connection between a destination and a source.

ADDRESSING WITH CLASSES

This addressing method divides the IP address into five separate classes based on four address bits.

HISTORY OF CLASSES

- - Classful addressing is a network addressing the Internet's architecture from 1980 till Classless Inter-Domain Routing was introduced in 1993.
 - Before 1980's :
 - When classes were not invented the 32 bit address were separated with fixed ratio of 8:24 where, 8 bits were allocated to network which means $2^8 = 256$ networks and 24 bits were allocated to hosts which means $2^{24} = 1,67,77,216$ hosts.
 - But the demands of people and companies changed during 1980's which gave raise to Class addressing.

CLASSFUL ADDRESSING:

□ As we saw there are basically 5 classes in IP addressing which are:

□ A class

□ B class

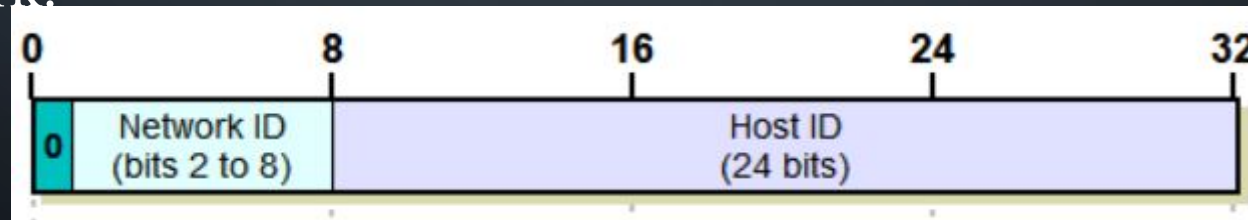
□ C class

□ D class

□ E class

CLASS A NETWORK

- This IP address class is used when there are a large number of hosts.
- In a Class A type of network, the first 8 bits (also called the first octet) identify the network, and the remaining have 24 bits for the host into that network.
- There are always specific predefined bits in a network.
- In class A there is 1 predefined bit i.e, 0 which makes network of class A 7 bit network.



CLASS A NETWORK

➤ How can we identify class A address :

- Range : We always take range from network of address, as in class A network starts with predefined 0. The first address is 00000000 and the last will be 01111111 which in decimal we can say 0 to 127 but the range of class A is 1.0.0.0 – 126.0.0.0 as 00000000 is never given to any network and 01111111 is used for research purpose. Eg. 12.0.0.0

➤ Networks :

- Number of networks : Class A has 8 bits assigned for network, since there is a predefined 0 at the beginning of network octate so there are 7 remaining free bits. Therefore the number of networks is $2^7 - 2 = 126$ networks. (for subtracting 2 there is same reason as range)

➤ Hosts :

- Number of hosts : Class A has 24 bits assigned for hosts i.e total number of hosts in class A comes out to be $2^{24} - 2 = 1,67,77,214$ hosts.

CLASS A NETWORK

- What we learnt in class A:

- Range 1.0.0.0 – 126.0.0.0

- Number of networks 126

- Number of hosts 1,67,77,214

- Subnet mask 255.0.0.0

- Example: 123.0.0.12, 100.0.0.34, 45.10.34.23

CLASS B NETWORK

- In a B class IP address, the binary addresses start with a predefined 2 bits 10.
- In this IP address, the class decimal number that can be between 128 to 191. The number 127 is reserved for loopback, which is used for internal testing on the local machine.
- The first 16 bits (known as two octets) help you identify the network. The other remaining 16 bits indicate the host within the network.



CLASS B NETWORK

➤ How can we identify class B address :

- Range : We always take range from network of address, as in class B network starts with predefined 10. The first address is 10000000 and the last will be 10111111 which in decimal we can say 128.0.0.0 to 191.255.0.0 Eg. 168.212.226.204

➤ Networks :

- Number of networks : Class B has 16 bits assigned for network, since there is a predefined 10 at the beginning of network so there are 14 remaining free bits. Therefore, the number of networks is $2^{14} - 2 = 16,382$ networks. (for subtracting 2 there is same reason as range)

➤ Hosts :

- Number of hosts : Class B has 16 bits assigned for hosts i.e total number of hosts in class B comes out to be $2^{16} - 2 = 65,534$ hosts.

CLASS B NETWORK

- What we learnt in class B:

- Range 128.0.0.0 - 191.255.0.0

- Number of networks 16,382

- Number of hosts 65,534

- Subnet mask 255.255.0.0

- Example: 132.12.45.1, 145.10.10.1, 130.54.1.2

CLASS C NETWORK

- ❑ In a C class IP address, the binary addresses start with a predefined 3 bits 110.
- ❑ In this IP address, the class decimal number that can be between 192 to 223.
- ❑ Mostly local area network used Class C IP address to connect with the network.
- ❑ The first 24 bits (known as three octets) help you identify the network. The other remaining 8 bits indicate the host within the network.



CLASS C NETWORK

➤ How can we identify class C address :

- Range : We always take range from network of address, as in class C network starts with predefined 110. The first address is 11000000 and the last will be 11011111 which in decimal we can say 192.0.0.0 to 223.255.255.0.

➤ Networks :

- Number of networks : Class C has 24 bits assigned for network, since there is a predefined 110 at the beginning of network so there are 21 remaining free bits. Therefore the number of networks is $2^{21} - 2 = 20,97,150$ networks. (for subtracting 2 there is same reason as range)

➤ Hosts :

- Number of hosts : Class B has 8 bits assigned for hosts i.e total number of hosts in class C comes out to be $2^8 - 2 = 254$ hosts.

CLASS C NETWORK

- What we learnt in class C:

- Range 192.0.0.0 - 223.255.255.0

- Number of networks 20,97,150

- Number of hosts 254

- Subnet mask 255.255.255.0

- Example: 199.12.45.1, 222.10.10.1, 210.54.1.2

CLASS D NETWORK

- Class D addresses are only used for multicasting applications. Class D is never used for regular networking operations. This class addresses the first three bits set to "1" and their fourth bit set to use for "0".
- All the values within the range are used to identify multicast groups uniquely.
- Therefore, there is no requirement to extract the host address from the IP address, so Class D does not have any subnet mask.



CLASS D NETWORK

□ What we learnt in class D:

□ Range: 224.0.0.0 to 239.255.255.255

□ Number of Networks: N/A

□ Number of Hosts per Network: Multicasting

□ Example: 225.12.45.1, 235.10.10.1, 22954.1.2

CLASS E NETWORK

- Class E IP address is defined by including the starting four network address bits as 1, which allows you to incorporate addresses from 240.0.0.0 to 255.255.255.255.
- However, E class is reserved, and its usage is never defined. Therefore, many network implementations discard these addresses as undefined or illegal.
- Example for a Class E IP address: 243.164.89.28



CLASS E NETWORK

- What we learnt in class E:

- Range: 240.0.0.0 to 255.255.255.255

- Number of Networks: N/A

- Number of Hosts per Network: Research/Reserved/Experimental

- Example: 250.12.45.1, 245.10.10.1, 242.54.1.2

LIMITATIONS OF CLASSFUL IP ADDRESSING

- Risk of running out of address space soon.
- Class boundaries did not encourage efficient allocation of address space.

CLASSLESS ADDRESSING:

- To reduce the wastage of IP addresses in a block, we use sub-netting. What we do is that we use host id bits as net id bits of a classful IP address. We give the IP address and define the number of bits for mask along with it (usually followed by a '/' symbol), like,
- 20.10.30.35 / 27 => Decimal
- 00010100.00001010.00011110.00100011 / 27 => Binary
- Total bits – Mask bits i.e 32 – 27
 - 5 bits
- So to identify 20.10.30.35 / 27 in which network this address lie , we will need first address and the last address of the network.
- So, first address will be : 00010100.00001010.00011110.001**00000** = 20.10.30.32
- And second address will be: 00010100.00001010.00011110.001**11111** = 20.10.30.63
- Therefore, we can say 20.10.30.35 / 27 IP lies in 20.10.30.32 network address.

SUBNETTING

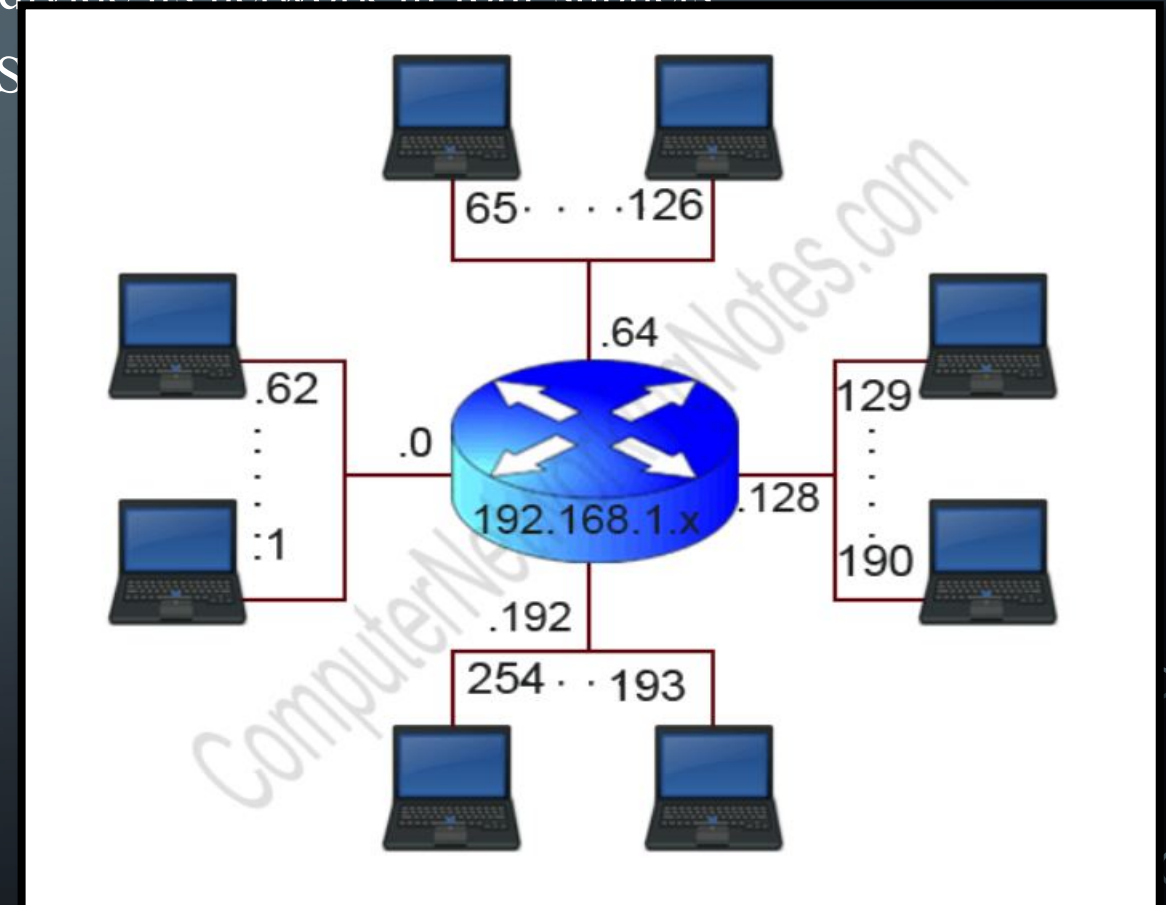
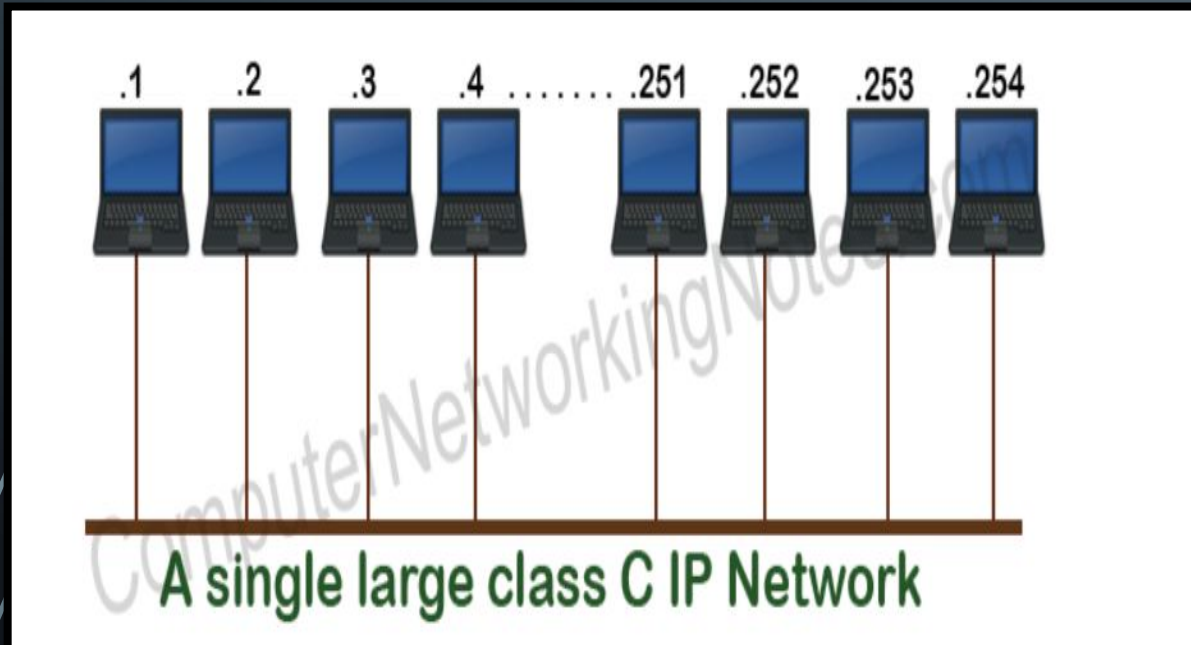
- Subnetting is a process of dividing a single large network in multiple smaller networks.
- In computer networking, Subnetting is used to divide a large IP network in smaller IP networks known as subnets.
- A default class A, B and C network provides 16,777,214, 65,534, 254 hosts respectively. Having so many hosts in a single network always creates several issues such as broadcast, collision, congestion, etc.

□ Let's take a simple example.

□ In a company there are four departments; sales, production, development and management. In each department there are 50 users. Company used a private class C IP network. Without any Subnetting, all computers will work in a single large network.

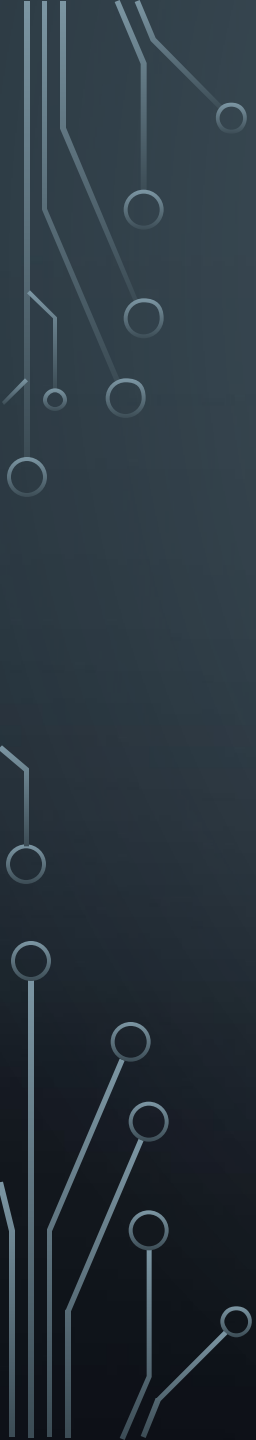
□ Since company has four departments, it can divide its network in four subnets

Following figure shows same network after S



SUPERNETTING:

- ❑ Supernetting is the opposite of Subnetting.
- ❑ In Supernetting, multiple networks are combined into a bigger network termed as a Supernetwork or Supernet.
- ❑ Supernetting is mainly used in Route Summarization, where routes to multiple networks with similar network prefixes are combined into a single routing entry, with the routing entry pointing to a Super network.
- ❑ More specifically, When multiple networks are combined to form a bigger network, it is termed as super-netting Super netting is used in route aggregation to reduce the size of routing tables and routing table updates.



□ Lets again take a simple example:

Class C block provides 256 host addresses

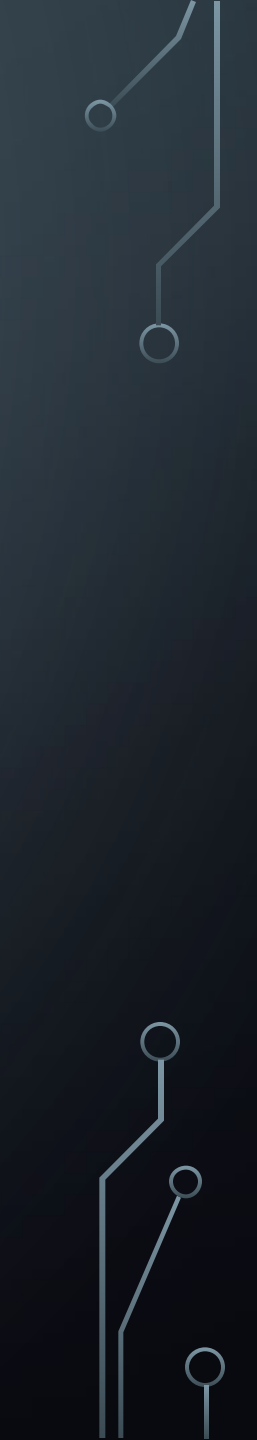
Class B block provides 65,500 host addresses

If any organisation wants 500 IP Addresses,

Class C block will not be sufficient.

Having 65,500 addresses in Class B block is waste of IP Addresses. To tackle with this, an organisation can apply for two contiguous Class C block of IP Addresses and combine them.

The combine network is called as supernet and the process of aggregation is called supernetting.



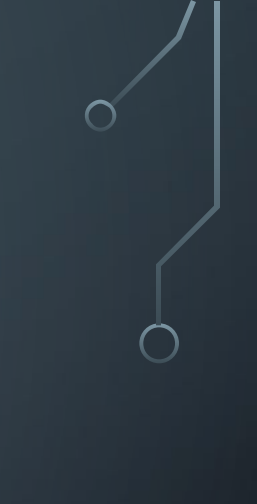
INTRO TO IPV6

- The depleting IPv4 addresses is one of the main reasons for a new IP version, IPv6.
- The size of an address in IPv4 address is 32-bit (4-bytes). This is increased much larger and the size of an address in IPv6 is 128 bits, which is four times longer than the 32-bit IPv4 address. The number of possible addresses in IPv4 is 2^{32} (4,294,967,296) but in IPv6 it is 2^{128} (3.4×10^{38}) addresses.
- This 32-bit large number is difficult to represent in binary format and therefore IPv4 addresses are represented in decimals, separated by a dot. An example of IPv4 address is 192.168.100.10.
- Hence the IPv6 addresses are represented in hexadecimal numbers, separated by a colon. An example of IPv6 address is 2001:0DB8:0000:0002:0022:2217:FF3B:118C.

DYNAMIC HOST CONFIGURATION PROTOCOL

□ What is DHCP?

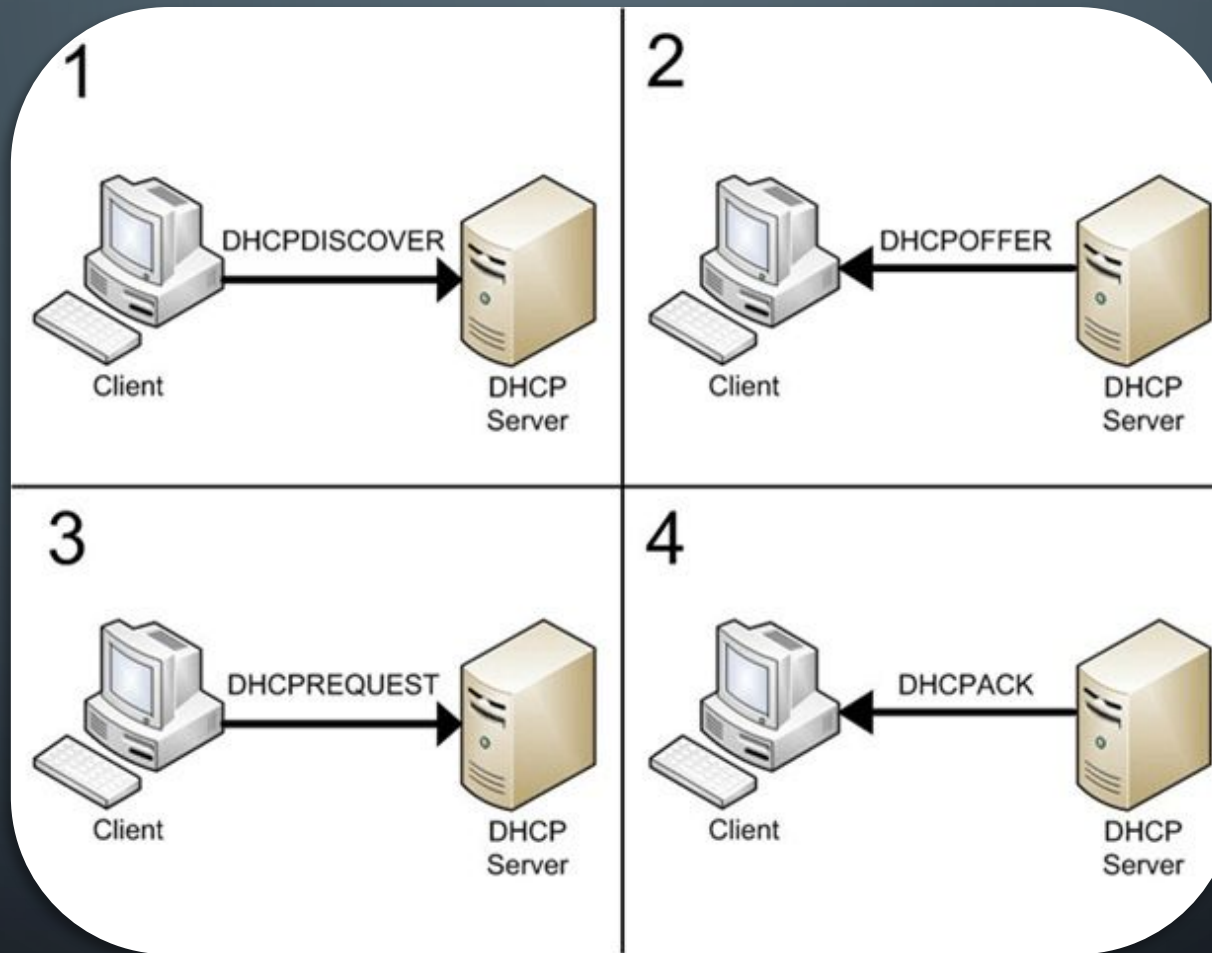
- DHCP automates and centrally manages these configurations. There is no need to manually assign IP addresses to new devices.
- DHCP can be implemented on local networks as well as large enterprise networks.
- DHCP is the default protocol used by the most routers and networking equipment.



FUNCTIONS OF DHCP:

- DHCP manages the provision of all the nodes or devices added or dropped from the network.
 - DHCP maintains the unique IP address of the host using a DHCP server.
- 
- 

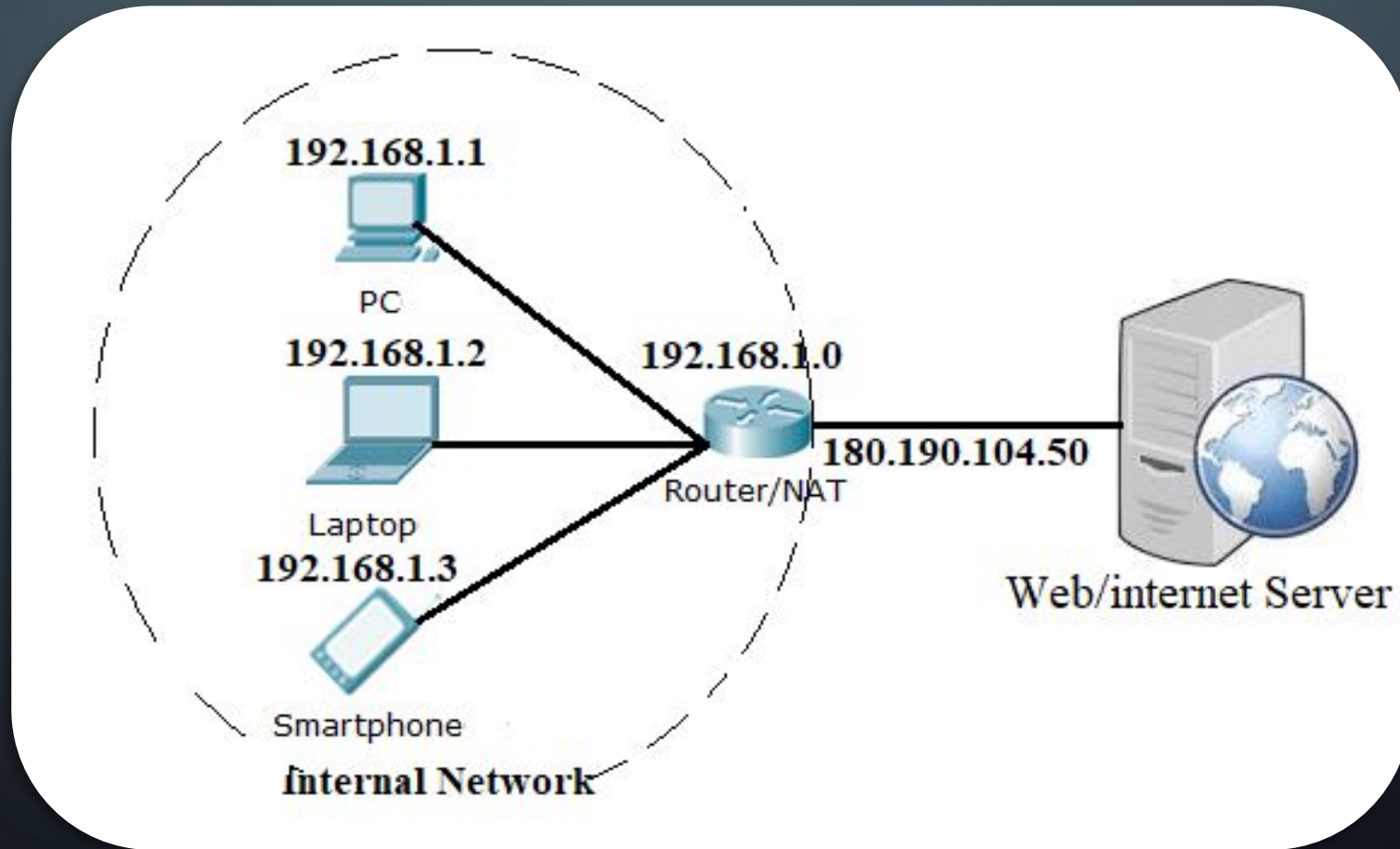
HOW DHCP WORKS?



NETWORK ADDRESS TRANSLATION (NAT)

- Network Address Translation (NAT) is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts.
- Also, it does the translation of port numbers i.e., masks the port number of the host with another port number, in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table.

WORKING OF NAT:





TYPES OF NAT:

□ Static NAT

□ Dynamic NAT

□ Port address translation NAT



□ ADVANTAGES OF NAT –

- NAT conserves legally registered IP addresses .
- It provides privacy as the device IP address, sending and receiving the traffic, will be hidden.
- Eliminates address renumbering when a network evolves.

□ DISADVANTAGES OF NAT –

- Translation results in switching path delays.
- Certain applications will not function while NAT is enabled.
- Also, router being a network layer device, should not tamper with port numbers(transport layer) but it has to do so because of NAT.





ETHERNET:

- Ethernet is the standard way to connect computers on a network over a wired connection.
- It provides a simple interface for connecting multiple devices, such as computers, routers, and switches.
- With a single router and a few Ethernet cables, you can create a LAN, which allows all connected devices to communicate with each other.

HISTORY

- Developed by Xerox PARC (Robert "Bob" Metcalfe)
- IEEE ratified (IEEE 802.3)
- 10 Mbps (Thicknet, Thinnet)
- 100 Mbps (Fast Ethernet)
- 1 Gbps (Gigabits Ethernet)
- 10 Gbps

ETHERNET SPECIFICATIONS:

Baseband and Broadband

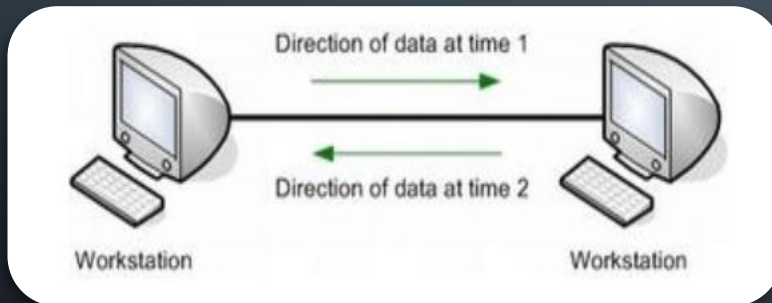
Name	IEEE Standard	Data Rate	Media Type	Maximum Distance
Ethernet	802.3	10 Mbps	10Base-T	100 meters
Fast Ethernet/ 100Base-T	802.3u	100 Mbps	100Base-TX 100Base-FX	100 meters 2000 meters
Gigabit Ethernet/ GigE	802.3z	1000 Mbps	1000Base-T 1000Base-SX 1000Base-LX	100 meters 275/550 meters 550/5000 meters
10 Gigabit Ethernet	IEEE 802.3ae	10 Gbps	10GBase-SR 10GBase-LX4 10GBase-LR/ER 10GBase-SW/LW/EW	300 meters 300m MMF/ 10km SMF 10km/40km 300m/10km/40km

*T-UTP *TX-UTP and STP *FX, SX, SR, SW-MMF *LX-MMF and SMF *LR/ER-SMF
LW/EW-SMF

HALF AND FULL DUPLEX ETHERNET

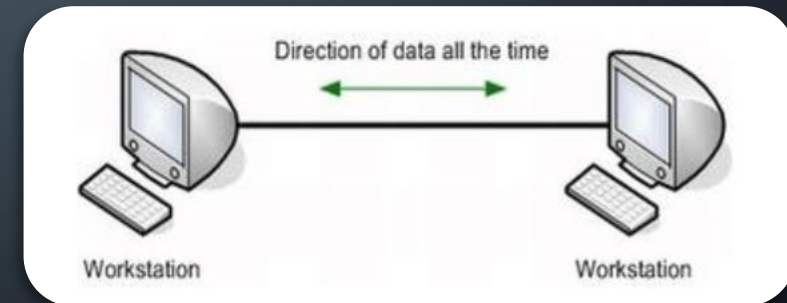
□ Half duplex:

- Half-duplex devices can only transmit in one direction at one time.
- Sending and receiving data simultaneously is not possible in half-duplex transmission.

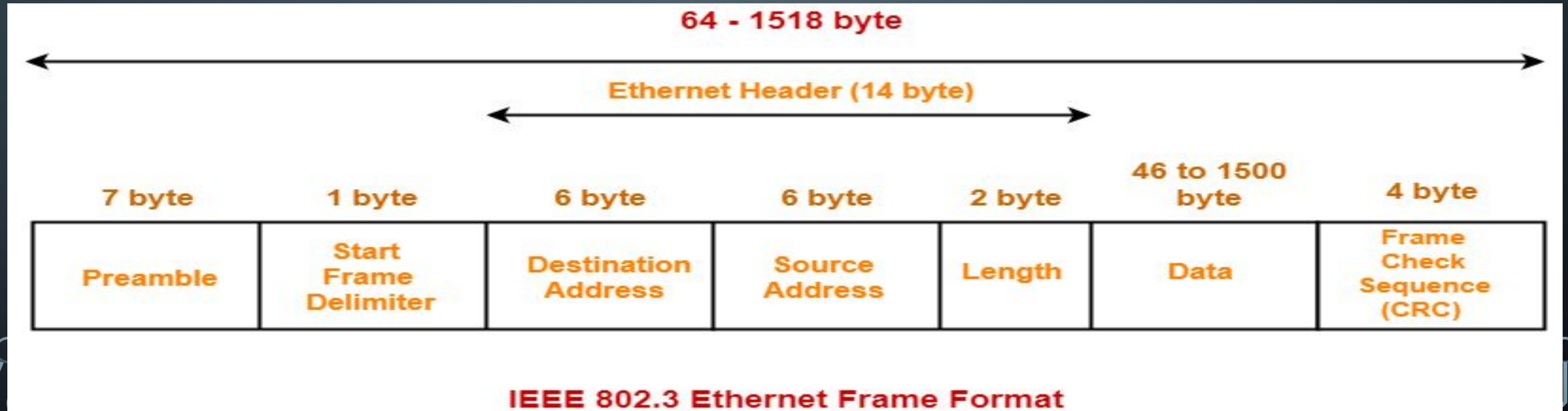


□ Full duplex:

- A full-duplex device is capable of bi-directional network data transmissions at the same time.
- To operate in Full-duplex mode, Ethernet requires at-least two pairs of wires.



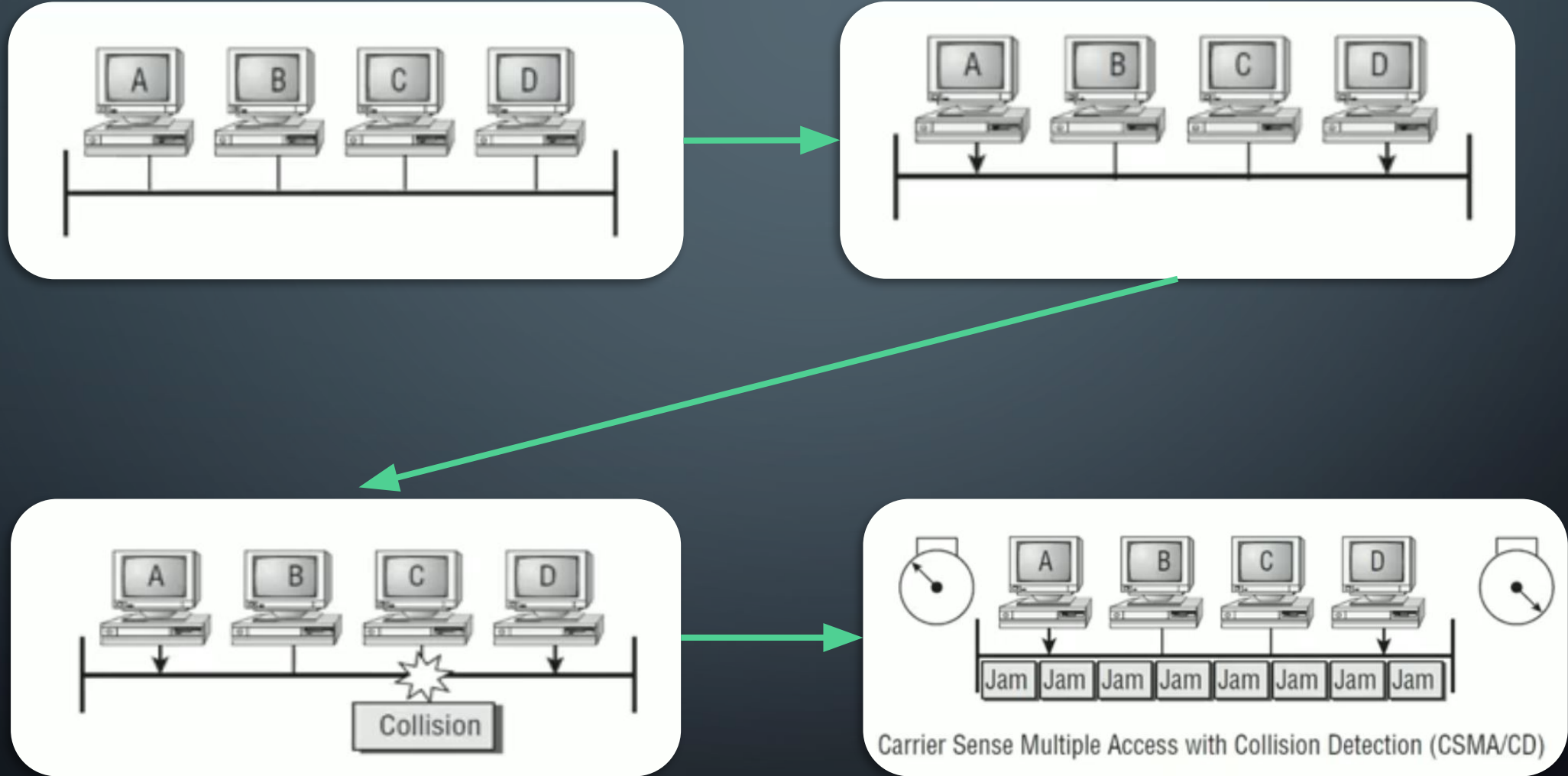
ETHERNET FRAME FORMAT:



PROPERTIES OF ETHERNET:

- Relative low cost and elegant implementation of LAN system.
- High flexibility
- High reliability
- Uses decentralized access control.
- Ethernet is a broadcast network.
- It does not provide any information to the source of the frame sent by it has successfully reached the destination.

CSMA/CD (CARRIER SENSE MULTIPLE ACCESS/ COLLISION DETECTION)



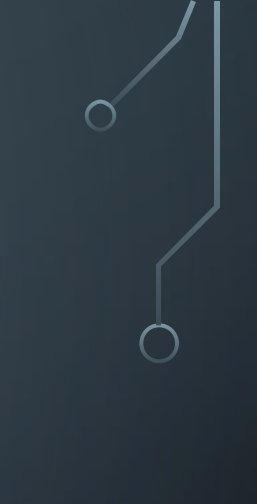
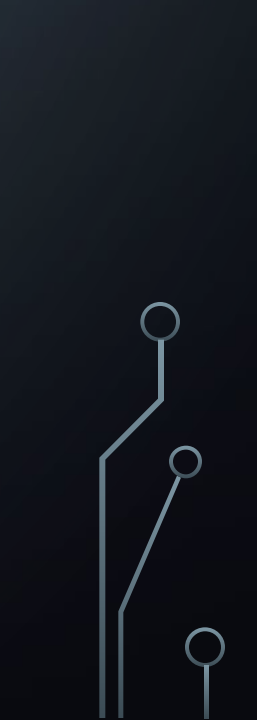
ETHERNET ADDRESSES

- The MAC address is often referred to as the Ethernet Address on an Ethernet network. They are in fact the same thing.
- The Ethernet address is 48 bits long and normally displayed as 12 hexadecimal digits.

3A-34-52-C4-69-B8



TYPES:

- Unicast Address
 - Multicast Address
 - Broadcast Address
- 
- 
- 

VIRTUAL LAN (VLAN)

□ Outcomes

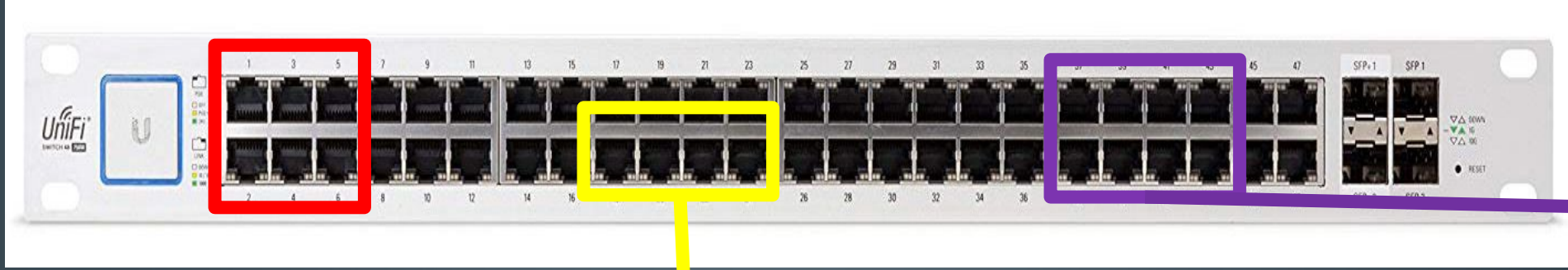
- Know the need for VLAN.

- Working of VLAN

- Benefits of VLAN

- Types of VLAN

- VLAN frame tagging



172.16.15.0



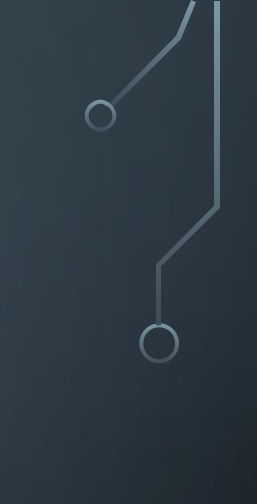
10.10.10.0



192.62.0.1

WORKING OF VLAN:

- VLAN is a logical partition of layer two network.
- Multiple partitions can be created allowing for multiple VLAN to co-exist.
- Each VLAN is a broadcast domain, usually with its own IP VLAN are mutually isolated and packets can only pass between them via a router.
- The partitioning of the layer 2 network takes place inside a layer 2 devices usually via switch
- The host grouped within a VLAN are unaware of the VLAN's existence.



BENEFITS OF VLAN:

- Security
 - Cost reduction
 - Shrinks broadcast domains
 - Improved IT staff efficiency
 - Simpler project and applications management
- 
- 

TYPES OF VLAN

□ Data vlan

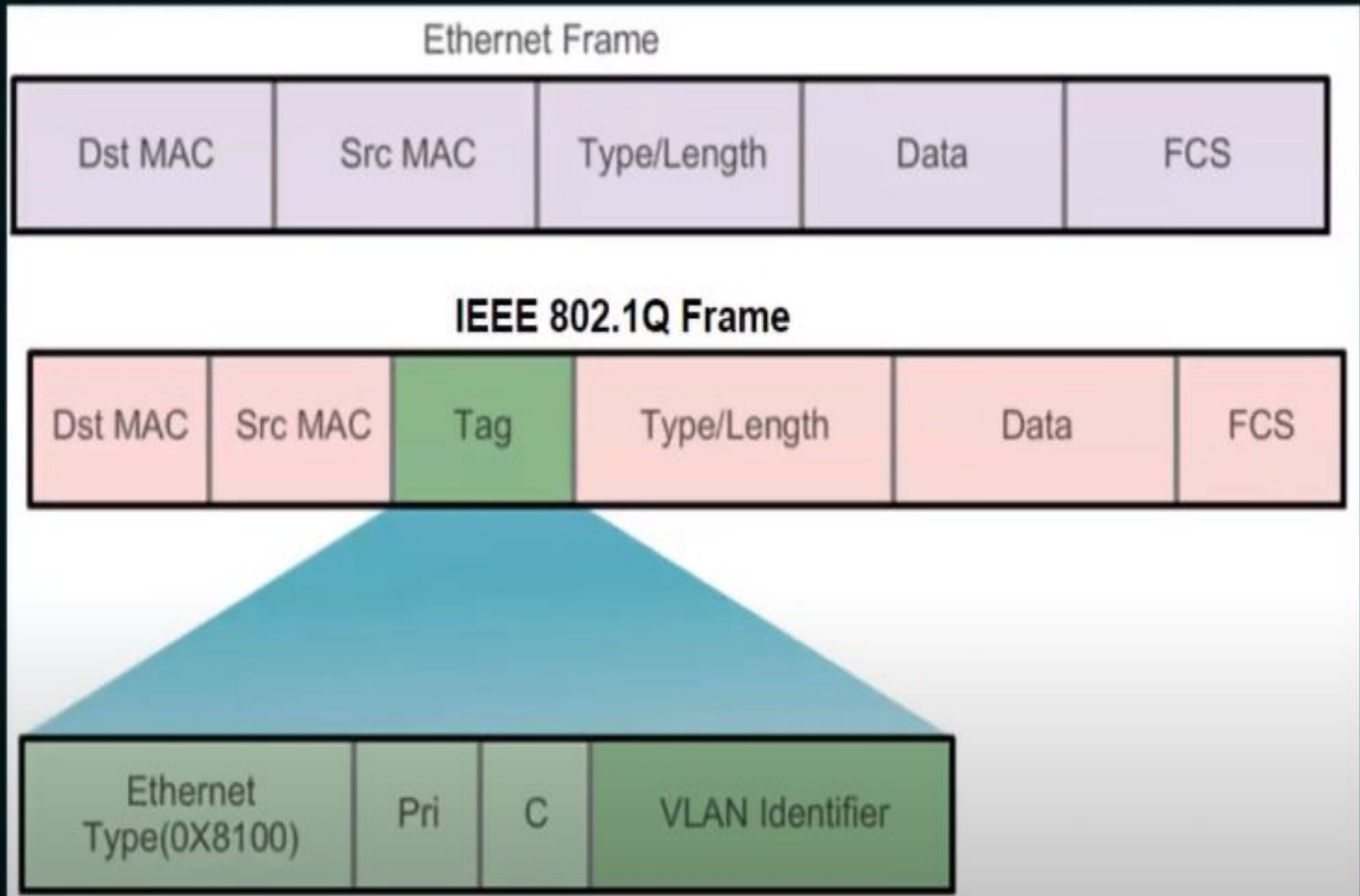
□ Default vlan

□ Native vlan

□ Management vlan

□ Voice vlan

VLAN FRAME TAGGING

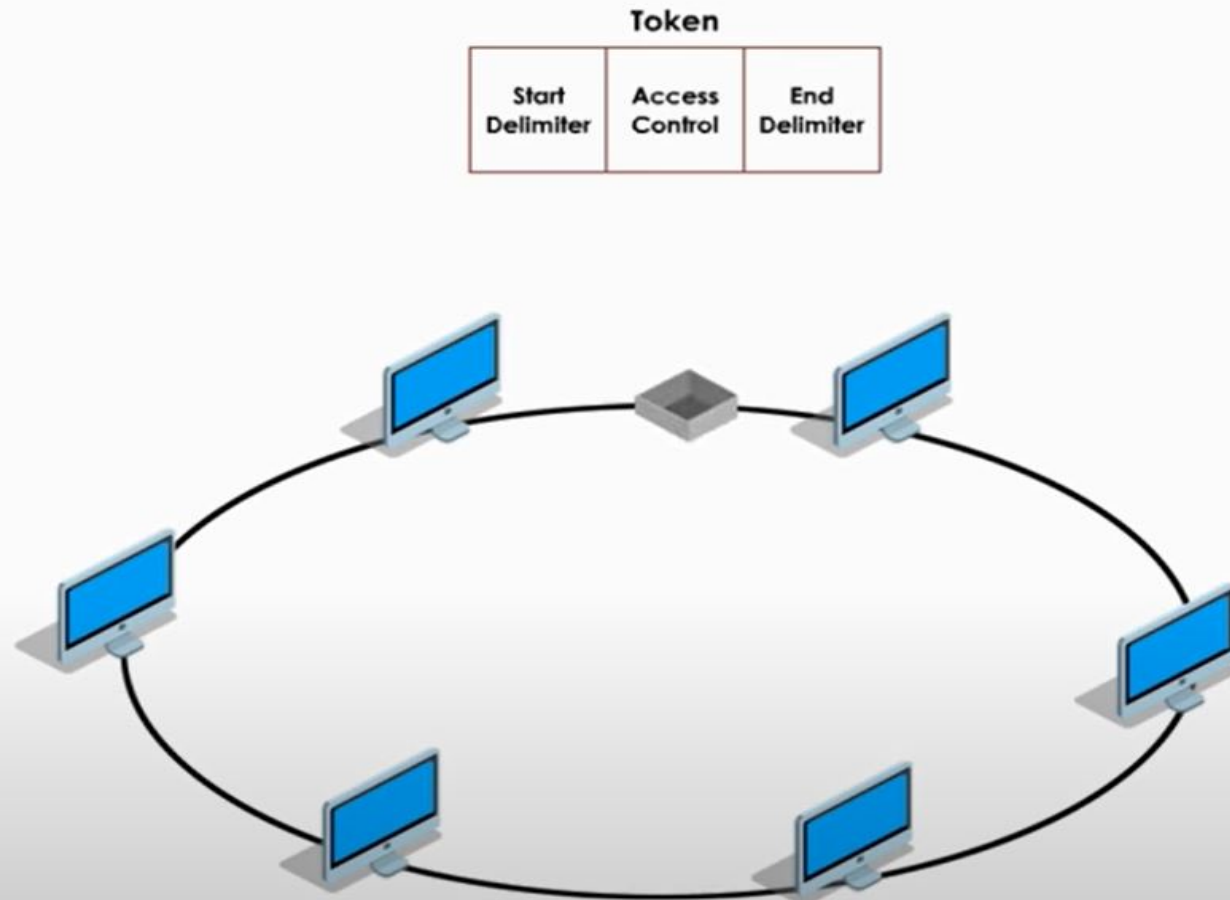


VLAN TAGGING

□ **Frame tagging**

- A frame tagging is the process of adding a vlan identification header to the frame.
- It is used to properly transmit multiple vlan frames through a trunk link
- Switches tags frames to identify the vlan to that they belong. Different tagging protocols exist, IEEE 802.1Q is a very popular example of it
- The protocol defines the structure of the tagging header added to the frame.
- Switches add vlan tags to the frames before placing them into trunk links and remove the tags before forwarding frames through non- trunk ports
- When properly tagged the frames can traverse any number of witches via trunk links and still be forwarded trunk links within the correct vlan at the destination

TOKEN RING



TOKEN RING

- Topology
- Access control
- Acknowledgment
- Encoding
- Data flow
- Data rates

LINKS OF YOUTUBE VIDEO

- Vlan: <https://www.youtube.com/watch?v=ez24W5oTU3U&t=445s>
- Token ring: https://www.youtube.com/watch?v=6r5j_ayCq5A
- You can watch his video series of token ring for more information .

THANK YOU!

